

# Ethics Application 2026 – Completed Draft

## Local Trait Inference on Smartphones

**DRAFT – NOT FOR SUBMISSION OR DATA COLLECTION**

Administrative and data-management fields marked “TO CONFIRM” require supervisor approval.

## Brief Description of the Research Project

### Principal Investigator

#### Academic title and name

**TO CONFIRM:** Principal investigator.

#### Research group

Science and Technology for Peace and Security (PEASEC); collaboration with Work and Engineering Psychology to be confirmed.

#### Department

Department of Computer Science, Technical University of Darmstadt; possible involvement of the Department of Human Sciences.

### Information on the Research Project

#### Working title

Local Trait Inference on Smartphones: User Perceptions of On-Device AI Profiling

#### Brief description (maximum 700 characters)

This bachelor-thesis study examines how adults perceive privacy-relevant profiles generated by an AI model directly on their Android phone. Participants select approximately 20 of their own images or PDF documents. Raw files remain on the device; the app displays probabilistic file-level inferences and a combined profile. Participants complete privacy, technology-affinity, demographic, per-inference, and final perception questions. Optional front-camera reaction tracking stores emotion labels but no images. The study evaluates expected capability, perceived accuracy, sensitivity, surprise, concern, and willingness to grant file access.

#### Planned project running time

26 May 2026 to 30 November 2026.

#### Estimated timeframe for data collection

1 September 2026 to 31 October 2026 at the latest, and only after a positive ethics vote and final approval of all study materials.

#### Other participating researchers

Atabey Tarik Akkum (student researcher, Bachelor of Science in Computer Science); Simon Althaus, M.Sc. (thesis supervisor, PEASEC); Dr. rer. nat. Nina Gerber (psychological and user-study consultation; formal role to be confirmed); **TO CONFIRM:** principal investigator and any additional data analysts.

#### The study is part of

☒ B.A. Thesis   ☐ M.A. Thesis   ☐ Dissertation   ☐ Course   ☐ Other

#### Motivation for the application

☒ The project touches on ethically relevant issues, specifically personal data, privacy-sensitive AI inferences, optional camera-based emotion classification, and the possible incidental presence of third-party information in participant-selected files.

## Checklist

| Question                                                                                        | Yes                                 | No                                  |
|-------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
| Physical stress or hazards?                                                                     | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Pain or mental stress through strong emotions beyond everyday life?                             | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Deception or manipulation of participants?                                                      | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Patients included?                                                                              | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Minors included?                                                                                | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Possible exposure of individuals, minorities, or vulnerable groups through statistical linking? | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Might third parties be involved or affected?                                                    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Is personal data collected from participants?                                                   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Hazardous substances used or produced?                                                          | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Publication without guaranteed participant anonymity?                                           | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Research with commercially created datasets?                                                    | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Uncertainty about data origin or lawful collection?                                             | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Commercial project partner also using collected data?                                           | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| In the applicants' opinion, is the study ethically uncritical?                                  | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |

**Checklist rationale.** The study does not intentionally induce distress beyond ordinary encounters with privacy notices or personalized digital services. Some participants may nevertheless experience concern, surprise, or mild discomfort after seeing probabilistic personal inferences. Personal data are collected. Third-party information could appear in selected files despite instructions; therefore this issue is conservatively marked “Yes.” The proposal is not classified as ethically uncritical because it deliberately studies privacy-sensitive profiling and includes optional camera-based processing.

# Application: Detailed Description of the Research Project

## Aim and research questions

Modern mobile devices can run multimodal AI locally. This allows an application to infer interests, activities, plans, education or occupational context, and other personal signals without transmitting the underlying images or documents. The study investigates:

1. what privacy-relevant information a local multimodal model can infer from participant-selected smartphone files;
2. how participants rate individual inferences in terms of unexpectedness, sensitivity, and perceived accuracy;
3. whether experiencing local profiling changes expectations about smartphone AI, concern about repeated background analysis, reassurance associated with the statement “data stays on the device,” and willingness to grant broad file access; and
4. how privacy concern and affinity for technology interaction relate to these perceptions.

The optional camera module provides exploratory emotion-label observations during study interaction. It is not used to identify participants or diagnose psychological or medical conditions.

## Design and participant tasks

The study uses a within-participant app-based design. Approximately 100 adults will use the Android research prototype on an Android smartphone. The planned procedure is:

1. Read the participant information and provide informed consent through unticked in-app checkboxes. Participation cannot proceed without the required consent. Optional front-camera tracking has a separate opt-in and is not required.
2. Complete demographic questions, IUIPC-8, the nine-item Affinity for Technology Interaction (ATI) scale, and four baseline questions about local AI expectations.
3. Select approximately 20 personally owned image files or PDF documents. The app instructs participants not to select files depicting identifiable third parties or containing third-party personal information. Files can be skipped and the study can be stopped at any time.
4. Analyze the files locally with Gemma 4 E2B. Images are analyzed directly. For PDFs, only a rendered preview of the first page is analyzed. The model produces structured probabilistic inferences; participants are explicitly told that these are guesses rather than confirmed facts.
5. After each successful inference, rate how unexpected, sensitive, and accurate it appears on five-point verbal scales. Completed inferences can be evaluated while later files are still being processed.
6. Generate a compact combined profile from the file-level outputs and answer six final questions about local AI capability, repeated background profiling, data-locality reassurance, undisclosed summary transmission, commercial use, and future file-access decisions.
7. Export one pseudonymized study record containing questionnaire responses, model outputs, ratings, technical metadata, optional emotion labels, and the combined profile. Raw selected files and camera images are not included.

## Participants, recruitment, setting, duration, and compensation

Inclusion criteria are age 18 years or older, capacity to provide informed consent, and ability to use the Android prototype. Minors, patients, and vulnerable groups are not specifically recruited. No medical or psychological condition is assessed.

**TO CONFIRM:** Recruitment route, study setting, and compensation. Current proposal: convenience sampling among adults accessible to the research team, without coercion and without recruitment from a direct dependency relationship. Confirm whether TU channels or participant pools will also be used and whether compensation or course credit is offered.

The active questionnaire and analysis procedure is expected to require approximately 25–40 minutes. A first-time model download of approximately 2.5 GB may add waiting time depending on the network and can run while baseline questions are completed. Wi-Fi is recommended. The participant-facing duration will be finalized after a short pilot.

## Potential benefits

Participants receive no guaranteed personal benefit. The study may increase understanding of current on-device AI capabilities. Scientifically, it contributes evidence about how local processing affects privacy expectations and whether avoiding raw-file upload is sufficient to reassure users when derived profiles can still be transmitted.

## Risks and burden

There are no physical risks. The principal foreseeable burden is mild concern, surprise, embarrassment, or discomfort caused by inaccurate or sensitive-seeming inferences. Mitigations are voluntary file selection, explicit uncertainty language, the ability to skip files or stop immediately, no requirement to select highly sensitive material, and contact details for questions or complaints. No deception is used. The true purpose of the study is disclosed before participation.

Participants are warned not to include images of other identifiable people or documents containing third-party personal information. If such a file is selected accidentally, it can be removed and will not be exported as a raw file. The final implementation will replace filenames and local URIs with random file identifiers before export.

## Optional front-camera reaction tracking

Participants may separately enable front-camera emotion tracking. If enabled, a low-resolution frame is sampled approximately every 12 seconds while the relevant study interface is active. Each frame is analyzed locally by a task-specific emotion-classification model. Only timestamp, study event, model identifier, predicted emotion label, and confidence are retained. No audio is recorded. Frame bytes and temporary camera files will be deleted immediately after each classification and will not be exported. Declining or stopping this feature has no effect on participation.

## Handling of Checklist Issues Marked “Yes”

**Personal data.** Questionnaire responses, consent decisions, per-inference ratings, model-generated inferences, a combined profile, technical metadata, and optional emotion labels are personal data. Data minimization, pseudonymization, restricted access, purpose limitation, and defined deletion periods will be used. Raw selected files remain local and are not exported.

**Potentially affected third parties.** A participant-selected file could contain another person’s image or information. Instructions prohibit such selection unless the third party has independently consented. Participants can remove files before analysis. No raw file is exported. Random file identifiers prevent filenames from disclosing third-party information.

**Ethical sensitivity.** The study deliberately makes privacy consequences visible. It avoids deception, coercion, diagnostic claims, and intentionally alarming language. Outputs are labelled as probabilistic. Participants can terminate without disadvantage.

## Regulations and Standards

The study follows the EU General Data Protection Regulation (GDPR), the Hessian Data Protection and Freedom of Information Act (HDSIG), informed-consent and data-minimization principles, the DFG Guidelines for Safeguarding Good Research Practice, and applicable TU Darmstadt research-data and information-security requirements. The study will begin only after ethics approval. The final data flow and storage arrangement will be reviewed with the TU Darmstadt Data Protection Officer if required.

# Data Protection

## Methods and data collection instruments

App-based experiment and questionnaires using an Android research prototype. Instruments comprise in-app consent, demographic items, IUIPC-8, ATI, custom baseline and final questions, per-inference verbal rating scales, local Gemma inference, technical runtime metadata, and optional local front-camera emotion classification.

## Affected groups and number

Approximately 100 adults aged 18 or older. No targeted recruitment of minors, patients, or vulnerable groups. Third parties are potentially affected only if a participant disregards the instruction not to select files containing other identifiable people or their personal data.

## How consent is obtained

Before any study data are collected, participants receive the full information sheet in the app and confirm required statements using initially unticked checkboxes. They consent to participation, local processing of selected files, and pseudonymized export of questionnaire and inference data. Camera tracking requires an additional optional opt-in. Consent and participation can be withdrawn without disadvantage.

## Types of personal data

Random study identifier; age group, gender including “prefer not to say,” and education category; IUIPC-8 and ATI responses; local-AI expectation items; consent decisions; per-inference unexpectedness, sensitivity, and accuracy ratings; structured model inferences and combined profile; model/runtime metadata and latency; optional timestamped emotion label and confidence. The final export will not contain raw files, camera frames, original filenames, local file paths, names, email addresses, or phone numbers.

## Deletion periods

a) Raw selected files are processed locally and are never copied to the research server. Rendered PDF previews and camera frames are deleted immediately after use or at latest when the local study session ends. b) **TO CONFIRM:** retention period for pseudonymized and anonymized research data; proposed distinction between working data and the data underlying thesis/publications. c) **TO CONFIRM:** retention and separate storage period for consent records. d) No audio is collected; temporary camera images are deleted immediately, and only derived labels are retained under the pseudonymous study ID.

## Pseudonymization/anonymization

The app assigns a random study ID unrelated to the participant’s identity. Exported files use randomized per-file identifiers. Recruitment contact details, if any, are stored separately and are never joined with study records. Analyses and publications use aggregated results. A participant can use the study ID to request deletion until the agreed anonymization cutoff.

## Local or online analysis and security

Raw-file and camera-frame analysis occurs locally on the Android phone. One JSON study record is transmitted only after consent using encrypted transport to **TO CONFIRM: the approved TU/PEASEC server and technical safeguards**. Stored records are pseudonymized, protected by role-based access, and included in institutional backup and deletion procedures. The production study will avoid third-party model downloads that disclose participant IP addresses, for example through TU-hosted distribution or supervised preinstallation.

## Persons analyzing the data and confidentiality

Atabey Tarik Akkum and the confirmed thesis supervisors/researchers will analyze the data. **TO CONFIRM: final named access list**. Access is limited to the study team. All researchers are bound by institutional confidentiality and data-protection obligations.

## Deadline for deletion requests

Participants can request deletion using their study ID until **TO CONFIRM: the date or processing milestone after which records are irreversibly anonymized and can no longer be linked to an individual participation**.

## Subsequent use

No subsequent use or publication of raw participant files or camera images is planned. Anonymized aggregate results may be used in the bachelor thesis and related scientific publications/presentations. **TO CONFIRM: whether pseudonymized record-level data may be retained for follow-up research; default recommendation: no**

reuse beyond the approved project without renewed legal and ethical review.

## Attachments

- ☒ Informed consent and participant information
- ☒ Questionnaires
- ☐ Research proposal
- ☐ Images/screenshots
- ☐ Other

## Signature

**TO CONFIRM:** The confirmed principal investigator must review and sign.

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Date and signature of the Principal Investigator

# Participant Information and Informed Consent

## Local Trait Inference on Smartphones: User Perceptions of On-Device AI Profiling

### DRAFT – NOT FOR SUBMISSION OR DATA COLLECTION

Administrative and data-management fields marked “TO CONFIRM” require supervisor approval.

## Information Sheet

Please read this information carefully before deciding whether to participate. You may ask questions at any time. Participation is voluntary.

### Who is conducting the study?

This study is conducted as part of Atabey Tarik Akkum’s bachelor thesis at Technical University of Darmstadt, Department of Computer Science, Science and Technology for Peace and Security (PEASEC).

**Student researcher:** Atabey Tarik Akkum, [AAkkum@outlook.com](mailto:AAkkum@outlook.com)

**Thesis supervisor:** Simon Althaus, M.Sc., [althaus@peasec.tu-darmstadt.de](mailto:althaus@peasec.tu-darmstadt.de)

**Principal investigator:** **TO CONFIRM:** name, institutional address, telephone number, and email

**Psychological/user-study contact:** Dr. rer. nat. Nina Gerber, [n.gerber@psychologie.tu-darmstadt.de](mailto:n.gerber@psychologie.tu-darmstadt.de) **TO CONFIRM:** confirm formal role and whether this contact should appear

### Purpose of the study

Modern smartphones can run powerful artificial-intelligence models directly on the device. These models may infer personal interests, activities, plans, education or occupational context, and other traits from locally stored images and documents even when the original files are not uploaded. This study examines what participants expect from this technology, how they perceive file-level and combined profile inferences, and whether local processing changes their privacy assessment.

The model outputs are probabilistic guesses. They may be correct, partly correct, or incorrect. They are not psychological, medical, financial, or legal assessments.

### Who can participate?

You may participate if you are at least 18 years old, can provide informed consent, and can use the Android study application. Please do not participate if you do not wish to select any personal files for local analysis.

### What will happen?

1. You read this information and provide consent in the app.
2. You answer demographic questions, questions about privacy concerns, affinity for technology interaction, and expectations about local smartphone AI.
3. You select approximately 20 of your own images or PDF documents. Select only files that you are comfortable using for this study. Do not select images of other identifiable people or documents containing another person’s personal information unless that person has independently consented.
4. The app analyzes each selected file directly on the phone. For PDF documents, only the first-page preview is analyzed. The app shows a short probabilistic inference after each file.
5. You rate how unexpected, sensitive, and accurate each inference appears.
6. The app combines the file-level results into one compact possible personal profile. You then answer final questions about local AI profiling and file access.
7. A pseudonymized JSON study record is transmitted or exported for research analysis. It contains your questionnaire responses, ratings, model-generated inferences, combined profile, consent decisions, and technical metadata. It does not contain the selected raw files or camera images.

### Duration and compensation

The active study is expected to take approximately 25–40 minutes. Installing the approximately 2.5 GB local AI model for the first time may add waiting time depending on the network, although baseline questions can be answered during the download. Wi-Fi is recommended.

**TO CONFIRM:** State compensation or explicitly state that no compensation/course credit is provided. Confirm the final duration after piloting.

#### Optional front-camera reaction tracking

You may separately choose to enable front-camera emotion tracking. If enabled, a low-resolution frame is sampled approximately every 12 seconds while the relevant study interface is active and analyzed directly on the phone. No audio is recorded. Camera frames are deleted immediately after classification and are not exported. The retained record contains only the timestamp, study event, model identifier, predicted emotion label, and confidence. The emotion label may be inaccurate and is not a diagnosis. You can disable this feature at any time without ending the study or losing compensation.

#### Possible benefits

There is no guaranteed direct benefit. You may learn more about the capabilities and privacy implications of local smartphone AI. The scientific results may help researchers design more transparent permissions, disclosures, and privacy protections for on-device AI systems.

#### Possible risks and discomfort

The study involves no physical risk beyond ordinary smartphone use. Some model inferences may feel inaccurate, sensitive, surprising, embarrassing, or uncomfortable. You control which files are selected and may skip a file, disable optional camera tracking, pause, or stop the study at any time. You are not required to choose highly sensitive files. The app clearly states that all inferred traits are uncertain guesses.

If you accidentally select a file containing third-party information, stop the analysis or remove the file. Raw files are not included in the research export.

#### Voluntary participation and withdrawal

Participation is voluntary. You may stop at any time without giving a reason and without disadvantage. Declining optional camera tracking does not affect participation. You may request access, correction, restriction, or deletion of your pseudonymized study record using your study ID until **TO CONFIRM: the deletion/anonymization cutoff**. Once records have been irreversibly anonymized and can no longer be linked to your study ID, individual deletion is no longer possible.

# Privacy Statement

Data processing for this study follows the General Data Protection Regulation (GDPR) and the Hessian Data Protection and Freedom of Information Act (HDSIG). Data are used only for the research purposes described here. Processing based on consent follows Article 6(1)(a) GDPR. You may withdraw consent with effect for the future.

## Data controller

Technical University of Darmstadt  
Karolinenplatz 5, 64289 Darmstadt, Germany  
Telephone: +49 6151 16-01  
<https://www.tu-darmstadt.de>

## Data collected

The study collects:

- a random study identifier;
- age group, gender category with a “prefer not to say” option, and highest education category;
- IUIPC-8 privacy-concern responses and ATI technology-affinity responses;
- baseline expectations and final opinions about local AI and privacy;
- per-file ratings of unexpectedness, sensitivity, and perceived accuracy;
- structured file-level inferences and the generated combined profile;
- model identifier/version, processing latency, timestamps, and non-identifying technical metadata; and
- if separately enabled, timestamped emotion labels and confidence values from local front-camera analysis.

The final study version will replace original filenames and local paths with random file identifiers. Names, email addresses, phone numbers, raw selected images/documents, rendered PDF previews, and camera frames are not included in the study record. Recruitment contact information, if required, is stored separately from study data.

## Local processing and transmission

Selected files and temporary camera frames are processed locally on the Android phone. Raw files and frames are not sent to the research server. After you have consented, the app exports one pseudonymized record containing questionnaire responses, model outputs, ratings, and technical metadata. Transmission will use encryption to **TO CONFIRM: the approved TU/PEASEC server and exact location**. The production study will distribute the AI model through an approved method that does not unexpectedly disclose participants’ IP addresses to a third-party model host.

## Confidentiality and access

Study records are stored under random identifiers. Any recruitment contact details are kept separately. Access is limited to Atabey Tarik Akkum and **TO CONFIRM: the final named principal investigator/supervisor access list**. Results in the thesis, presentations, or publications are reported in anonymized and aggregated form. No individual participant will be named.

## Storage and deletion

Raw selected files are not copied to the research server. Temporary previews and camera frames are deleted immediately after processing or at latest when the local study session ends. Pseudonymized study data are stored on **TO CONFIRM: approved server/location** until **TO CONFIRM: retention date or period**. Consent records are stored separately until **TO CONFIRM: retention date or period**. After the deletion period, identifiable or pseudonymized records are securely deleted; only irreversibly anonymized aggregate results may remain.

## Your rights

You have the right to request information about your data, correction, deletion, restriction of processing, objection where applicable, withdrawal of consent, and data portability where applicable. Withdrawal does not affect the lawfulness of processing performed before withdrawal.

For study questions or withdrawal, contact the principal investigator listed above and provide your study ID.  
For data-protection questions, contact:

Technical University of Darmstadt, Data Protection Officer  
Karolinenplatz 5, 64289 Darmstadt  
[datenschutz@tu-darmstadt.de](mailto:datenschutz@tu-darmstadt.de)  
<https://www.tu-darmstadt.de/datenschutz>

You may also lodge a complaint with:

The Hessian Commissioner for Data Protection and Freedom of Information  
Wilhelmstrasse 7, 65185 Wiesbaden  
Telephone: +49 611 1408-0  
[poststelle@datenschutz.hessen.de](mailto:poststelle@datenschutz.hessen.de)

## Declaration of Consent for the App-Based Study

For an app-based or online study, consent is given through initially unticked checkboxes. Participants must be able to retain or download this information sheet. Required consent statements:

- ☐ I confirm that I am at least 18 years old.
- ☐ I have read and understood the participant information and privacy statement and had the opportunity to ask questions.
- ☐ I voluntarily agree to participate and understand that I may stop at any time without giving a reason and without disadvantage.
- ☐ I agree that files I deliberately select may be analyzed locally on this phone for this study.
- ☐ I confirm that I will select only my own files and will not deliberately select files containing identifiable third parties or their personal information without their consent.
- ☐ I agree that my questionnaire responses, consent decisions, ratings, model-generated inferences, combined profile, and technical metadata may be stored and analyzed under a pseudonymous study ID for the purposes described above.
- ☐ I understand that selected raw files are not part of the research export and that the model outputs are probabilistic guesses, not confirmed facts or diagnoses.

Separate optional statement, not required for participation:

- ☐ I voluntarily agree to optional front-camera reaction tracking as described above. I understand that camera frames are analyzed locally and deleted immediately, while derived emotion labels, confidence values, timestamps, and study events are retained. I can disable this feature at any time.

**Study ID shown to participant:** \_\_\_\_\_

*The final participant-facing version must contain no red placeholders and must match the implemented app, server configuration, retention policy, and ethics-approved procedure exactly.*

# Questionnaire and Study Measures

## Local Trait Inference on Smartphones

### **DRAFT – NOT FOR SUBMISSION OR DATA COLLECTION**

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This appendix lists all participant-facing questionnaire items planned for the Android study. Scale values are stored numerically together with their verbal labels. The optional camera module is a data-collection instrument, not a questionnaire.

## A. Eligibility and Demographic Questions

**A1. Age confirmation** I confirm that I am at least 18 years old.

*Options: Yes / No. A “No” response ends eligibility.*

**A2. Age group** Which age group are you in?

*Options: 18–24; 25–34; 35–44; 45–54; 55 or older; prefer not to say.*

**A3. Gender** How do you describe your gender?

*Options: woman; man; non-binary; self-description; prefer not to say.*

**A4. Highest completed education level** What is the highest level of education you have completed?

*Options: school qualification; vocational training; bachelor’s degree; master’s degree or higher; other; prefer not to say.*

**Implementation alignment note:** The current prototype lacks “prefer not to say” for age, and lacks “self-description”/“other” options in two items. These options should be aligned before data collection.

## B. Internet Users’ Information Privacy Concerns – IUIPC-8

**Instruction:** The following statements concern online privacy. Please indicate how strongly you agree or disagree with each statement.

*Response scale: 1 strongly disagree; 2 disagree; 3 somewhat disagree; 4 neither agree nor disagree; 5 somewhat agree; 6 agree; 7 strongly agree.*

### Control

**B1.** Consumer online privacy is really a matter of consumers’ right to exercise control and autonomy over decisions about how their information is collected, used, and shared.

**B2.** Consumer control of personal information lies at the heart of consumer privacy.

### Awareness

**B3.** Companies seeking information online should disclose the way the data are collected, processed, and used.

**B4.** A good consumer online privacy policy should have a clear and conspicuous disclosure.

### Collection

**B5.** It usually bothers me when online companies ask me for personal information.

**B6.** When online companies ask me for personal information, I sometimes think twice before providing it.

**B7.** It bothers me to give personal information to so many online companies.

**B8.** I am concerned that online companies are collecting too much personal information about me.

**Scoring note for researchers:** Preserve the three dimensions Control, Awareness, and Collection. Confirm the planned scoring and psychometric analysis with the psychology supervisor before preregistration or final analysis.

**Source:** Gross, T. (2023). Toward Valid and Reliable Privacy Concern Scales: The Example of IUIPC-8. In *Human Factors in Privacy Research*. The eight-item form omits ctrl3 and awa3 from IUIPC-10 and uses a seven-point agreement scale.

## C. Affinity for Technology Interaction – ATI

Instruction: In the following questionnaire, we will ask you about your interaction with technical systems. The term “technical systems” refers to apps and other software applications, as well as entire digital devices such as a mobile phone, computer, TV, or car navigation system.

*Response scale: 1 completely disagree; 2 largely disagree; 3 slightly disagree; 4 slightly agree; 5 largely agree; 6 completely agree.*

- C1. I like to occupy myself in greater detail with technical systems.
- C2. I like testing the functions of new technical systems.
- C3. I predominantly deal with technical systems because I have to.
- C4. When I have a new technical system in front of me, I try it out intensively.
- C5. I enjoy spending time becoming acquainted with a new technical system.
- C6. It is enough for me that a technical system works; I do not care how or why.
- C7. I try to understand how a technical system exactly works.
- C8. It is enough for me to know the basic functions of a technical system.
- C9. I try to make full use of the capabilities of a technical system.

**Scoring note for researchers:** Items C3, C6, and C8 are reverse-coded. Compute the mean across all nine items after reversal. Report the mean, standard deviation, and internal consistency as appropriate.

**Source:** Franke, T., Attig, C., & Wessel, D. (2019). A Personal Resource for Technology Interaction: Development and Validation of the Affinity for Technology Interaction (ATI) Scale. *International Journal of Human-Computer Interaction*, 35(6), 456–467. <https://doi.org/10.1080/10447318.2018.1456150>

## D. Baseline Local-AI Expectations

Instruction: Please indicate how strongly you agree or disagree before seeing any file analysis.

*Response scale: 1 strongly disagree; 2 disagree; 3 neither agree nor disagree; 4 agree; 5 strongly agree.*

- D1. I expect a phone to infer sensitive information from selected local files.
- D2. The statement “data stays on the phone” makes this feel privacy-friendly.
- D3. I usually enjoy trying new digital technologies.
- D4. I am concerned about how apps can use information stored on my phone.

**Design note:** D3 overlaps conceptually with ATI and D4 overlaps with IUIPC-8. Confirm whether they are needed as brief face-valid items or should be removed to reduce burden and avoid redundant measurement.

## E. Post-Inference Questions

These three questions are shown after each successfully analyzed image or PDF first-page preview. The displayed probabilistic inference remains visible while the participant responds.

*Response scale: 1 not at all; 2 slightly; 3 moderately; 4 strongly; 5 very strongly.*

- E1. How unexpected was this inference?
- E2. How sensitive does this inference feel?
- E3. How accurate does this inference seem?

Each response is linked to a randomized file identifier and its model output. Original filenames and local paths are excluded from the final research export.

## F. Combined Profile and Final Questions

Before these questions, participants see one compact probabilistic personal profile synthesized from the file-level inferences. The following framing text is displayed:

Imagine a normal app performed this kind of local analysis regularly in the background and only sent inferred summaries or profile categories away.

*Response scale: 1 strongly disagree; 2 disagree; 3 neither agree nor disagree; 4 agree; 5 strongly agree.*

**F1.** Before this study, I expected my phone to be able to run this kind of AI analysis locally.

**F2.** I would be concerned if an app repeated this analysis regularly on local photos, screenshots, or documents.

**F3.** The statement “the original files never leave the device” would still reassure me if inferred summaries or profile categories were sent to a server.

**F4.** I would be comfortable if an app analyzed local files in the background and sent inferred summaries to a server without clearly notifying me each time.

**F5.** I would be comfortable if this inferred profile were used commercially, for example for advertisements, product offers, or sharing with another company.

**F6.** After seeing this profile, I would still allow broad photo or file access to an app.

## G. Optional Front-Camera Reaction Measure

This is not a self-report questionnaire. After separate opt-in, the app may sample a low-resolution front-camera frame approximately every 12 seconds while the relevant study interface is active. Local classification yields:

- timestamp;
- study-screen event;
- model identifier/version;
- predicted label from angry, disgust, fear, happy, neutral, sad, or surprise; and
- model confidence.

No audio is captured. The final implementation must delete each temporary frame immediately and retain no image bytes or file path. This measure is exploratory and must not be interpreted as a direct measure of the participant’s true emotional state or as a diagnosis.

## H. End-of-Study Information

Participants are reminded that:

- all displayed traits are probabilistic model outputs and may be wrong;
- the original selected files were processed locally and were not part of the research export;
- local processing does not by itself prevent privacy impact if derived summaries or profiles are transmitted; and
- they may contact the study team with questions or request deletion using their study ID until the stated cutoff.